

Ray Zhong

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EDUCATION

University of Michigan, Ann Arbor <i>Master of Science in Environment and Sustainability, Sustainable Energy Systems Track</i> • Relevant Coursework: Power Systems Market and Optimization, Machine Learning in Climate Science, Industrial Ecology	Ann Arbor, MI Aug 2025 - May 2027
Duke University and Duke Kunshan University (DKU), Dual Degree Program <i>Bachelor of Science in Environmental Science, Biogeochemistry Track</i> • Relevant Coursework: Biogeochemistry, Intro to GIS, Sustainable Investments, Intro to LCA, Energy and Environment	Durham, US Kunshan, China Aug 2021 - May 2025

PROFESSIONAL EXPERIENCES

Micron Technology <i>Global EHSS, Sustainability Intern</i> • Developed an end-to-end agentic AI automation pipeline within the Microsoft Copilot Studio environment to autonomously ingest sustainability reports, extracted and structured ~12000 data records, achieved 97.2% extraction accuracy. • Built a multi-agent workflow integrating automated validation, duplicate detection, exception handling, and dashboard visualization to transform unstructured supplier disclosures into analytics-ready sustainability insights for procurement.	Boise, ID May 2026 - Present
National Laboratory of the Rockies (Former NREL) <i>Graduate Researcher</i> • Engineered stochastic input parameters by processing ~600 empirical datasets regarding municipal landfill costs and constraints for end-of-life (EOL) PV modules, successfully integrating them into NREL's predictive energy models • Quantified environmental impacts of PV module disposal treatments through data modeling, calculating carbon footprints to provide validations for circular economy frameworks and support stakeholders' decision-makings with policy analysis	Ann Arbor, MI Jan 2026 - Present
Initiative for Sustainable Investment <i>Assistant Intern</i> • Compiled 25 structured inventories of eco-friendly technologies adopted by ~75 electronic information manufacturing enterprises, supporting the transition of traditional manufacturing toward low-carbon and sustainable practices • Reviewed and analyzed ESG and supply chain annual reports from ~200 enterprises in the renewable energy sectors, using a self-designed quantitative evaluation framework to assess key sustainability indicators (Scope 1 & 2 emissions)	Kunshan, China Apr 2024 - Jan 2025
CECEP Hundred Technical Service (Beijing) Co, Ltd <i>Analyst Intern, Green Consulting (ESG) Department</i> • Developed carbon abatement and cost-benefit financial models with Excel and R to evaluate the effectiveness of green loan financing across ~3,000 energy industry cases, contributing to Q2 2023 ESG impact reporting and portfolio-level assessment • Conducted research and synthesized green and transition finance policies into concise regulatory briefings, improving stakeholder accessibility to ESG-related requirements. Visualized the results with R and Excel	Beijing, China June 2023 - Aug 2023

PROJECTS

Solar Power Generation Forecasting with LSTM <i>Research Project, University of Michigan</i> • Built a CNN-LSTM hybrid machine learning solar PV forecasting model using high-resolution weather data (~200k data points), covering full data preprocessing and time-series modeling of two power plants' generation results • Achieved strong trend prediction performance with Pearson correlation ≈ 0.94 and $R^2 \approx 0.89$, and analyzed generalization and overfitting behavior using validation loss and residual diagnostics. Reached 78% general prediction curve match rate	Ann Arbor, MI Aug 2025 - Dec 2025
Spatial Analysis of Land-Use Change and Urban Climate Resilience <i>Undergraduate Thesis, Duke Kunshan University</i> • Processed 20 years of high-resolution Landsat satellite imagery using QGIS, the InVEST model and AI-driven convolutional LSTM model to evaluate the impact of land-use transformation on regional Urban Heat Island Effect across a 928 km ² area • Developed a Patch Cohesion Index (PCI) and conducted multiple linear regressions in R to quantify correlations between indicators (e.g. infrastructure expansion, land-cover fragmentation) and proposed strategic land-use optimizations	Wuhan, China Nov 2024 - Mar 2025

SKILLS & LEADERSHIP

Programming and Tools: Python, R, SQL (MySQL), Matlab, Simulink
Sustainability Software & Packages: OpenLCA, SimaPro, QGIS, InVEST, Microsoft Office